

Systems Analysis and Design Chapter

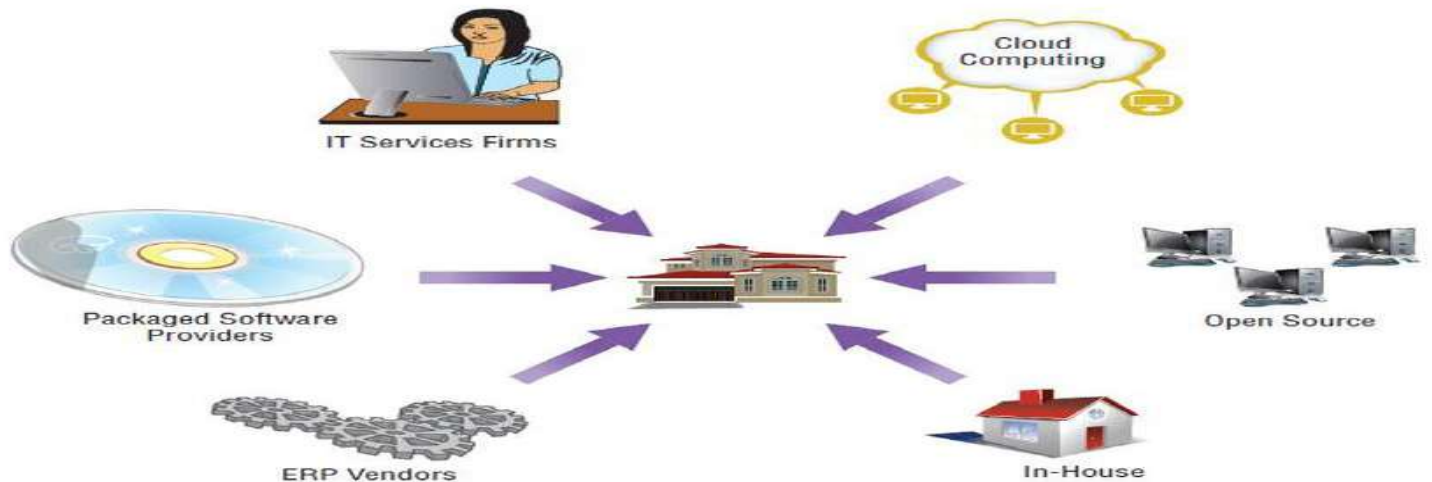
2: Sources of Software

Systems Acquisition

Outsourcing

- The practice of turning over responsibility for some or all of an organization's information systems applications and operations to an outside firm.

Six Sources of Software



Sources of Software cont.

1. IT Services Firms.

When

- does not have the expertise or the personnel to develop the system **in-house**.
- a suitable **off-the-shelf** system is not available.

2. Packaged Software.

- Problem Some off-the-shelf software systems cannot be modified to meet the specific, individual needs of a particular organization.

3. Enterprise Resource Planning (ERP) Systems

- A system that integrates individual **traditional business functions into a series of modules** so that a single transaction occurs seamlessly within a **single information system** rather than several separate systems such as accounting, distribution, manufacturing, or human resources.

Advantages

- A firm can integrate all parts of a business process in a unified information system.
- Flexible because additional modules can be added as needed once the basic system is in place.

Disadvantages

- The systems are **complex**.
- implementation can take a long time.
- Organizations do not have the expertise in-house to implement the systems, so they must rely on consultants of the software vendor, which can be expensive.
- In some cases, organizations must **change how they do business** in order use enterprise solutions.

The most known ERP vendors are **SAP** and **Oracle** which, control about 42 percent of the ERP market.

4. Cloud Computing

The provision of computing resources, including applications, **over the Internet**, so customers do not have to invest in the computing infrastructure needed to run and maintain the resources.

Cloud computing includes many areas of technology, including **software as a service** (often referred to as **SaaS**), which includes **Google Apps** and **Salesforce.com**, and **hardware as a service**, which allows companies to order server capacity and **storage** on demand.

Advantages

- Freeing internal IT staff.
- Gaining access to applications faster than via internal development.
- Achieving lower-cost access to corporate-quality applications.
- Through a cloud also makes it easier to walk away from an unsatisfactory systems solution.

Disadvantages

- The primary concern **is reliability**, but other concerns include **security** and compliance with government regulations.

5. Open-Source Software

- is freely available—not just the final product, but the **source code itself**.
- It is developed by a community of interested people instead of employees of a particular company.
- Open-source software performs the same functionality as functions as commercial software.
- **Examples are: Linux, MySQL, PostGresQL.**

SourceForge.net : Is a web site for organising and sharing open source SW.

How they can make money

- By providing maintenance and other services.
- By providing one version of the software for free and selling a more **fully featured version**.

6. In-House Development: Developing the software **inside** the company.

The company should have the required expertise in-house to choose that option.

- Hybrid solutions involving some purchased and some in-house software components are common.
- Some in-house software components are reused

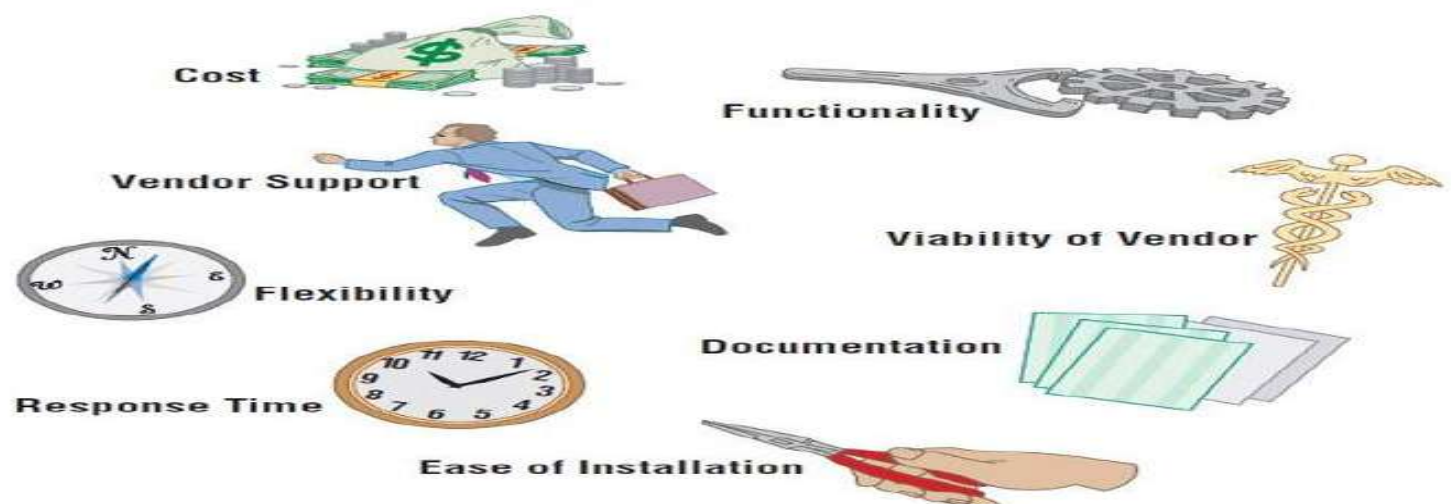
Some SW Companies

Specialization	Example Firms or Websites
IT Services	Accenture Capgemini Sogenti Computer Sciences Corporation (CSC) IBM HP
Packaged Software Providers	Microsoft Intuit Oracle Symantec Adobe
Enterprise Software Solutions	Oracle SAP AG
Cloud Computing	Amazon.com Google Microsoft Salesforce.com
Open Source	SourceForge.net

Comparison of Six Different Sources of Software Components

Producers	When to Go to This Type of Organization for Software
IT services firms	When task requires custom support and system can't be built internally or system needs to be sourced
Packaged software producers	When supported task is generic
Enterprise solutions vendors	For complete systems that cross functional boundaries
Cloud computing	For instant access to an application; when supported task is generic
Open-source software	When supported task is generic but cost is an issue
In-house developers	When resources and staff are available and system can be built from scratch

Choosing Off-the-Shelf Software



Choosing Off-the-Shelf

Software cont.

- **Vendor support and vendor viability** are the most important criteria.
- ranking the importance of the remaining criteria depends primarily on your specific situation.
- **Cost** involves comparing the cost of developing the same system in-house to the cost of purchasing or licensing the software package. Be sure to include a comparison of the cost of purchasing vendor upgrades or annual license fees with the costs you would incur to maintain your own software.
- **Functionality** refers to the tasks the software can perform and the mandatory, essential, and desired system features.

Questions to ask.

- Can the software package perform all, or just some, of the tasks your users need?
- If some, can it perform the necessary core tasks?
- **vendor support** refers to whether the vendor can provide support, and how much. Support includes assistance to install the software, to train user and systems staff on the software, and to provide help as problems arise after installation.
- Related to support is the **vendor's viability**. This latter point should be considered, it means vendor credibility.
- **Flexibility** refers to how easy it is for you, or the vendor, to customize the software. If the software is not sufficiently flexible, your users may have to adapt the way they work to fit the software.
- **Documentation** includes the user's manual as well as technical documentation. How understandable and up to date is the documentation? What is the cost for multiple copies, if required?
- **Response time** refers to how long it takes the software package to respond to the user's requests in an interactive session.
- Another measure of time would be how long it takes the software to complete running a job.
- **Ease of installation** is a measure of the difficulty of loading the software and making it operational.

Validating Purchased Software Information

- When you decide to outsource, Issue a **request for proposal (RFP)** to vendors.
- The RFP is a document provided to vendors to ask them to propose hardware and system software that will meet the requirements of a new system.
- Issuing an RFP gives you the opportunity to have vendors conduct the research you need in order to decide among various options.
- You can request that each bid submitted by a vendor contain information essential for you to decide on what best fits your needs.

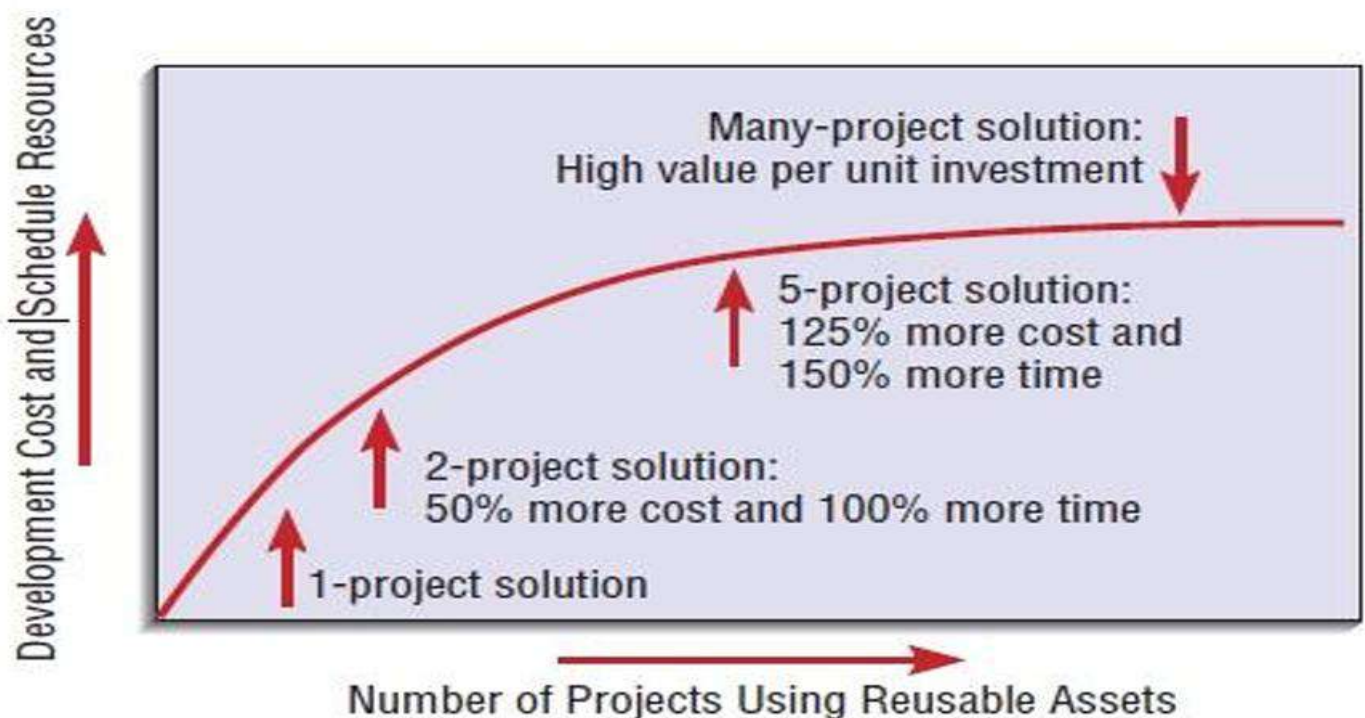
Software Reuse

- The use of previously written software resources, especially objects and components, in new applications.

Issues related to Reuse

- **Technical issues** include the current lack of a methodology for creating and clearly defining and labelling reusable components for placement in a library.
- **Organizational issues** include the lack of commitment to reuse, as well as the lack of proper training and rewards needed to promote it, the lack of organizational support reuse, and the difficulty in measuring the economic gains from reuse.
- Software reuse has **three basic steps**: abstraction, storage, and recontextualization.
- **Abstraction** involves the design of a reusable piece of software, starting from existing software assets or from scratch.
- **Storage** involves making software assets available for others to use.
- The problem is not simply putting software assets on a shelf; the problem is correctly labelling them.
- **Recontextualization**—or making the reusable asset understandable to developers who want to use it in their systems.

Investments necessary to achieve reusable components.



Approaches to Reuse

- **ad hoc**: is a reuse approach where, **individuals** are free to find or develop reusable assets on their own, but few, if any, organizational rewards are offered for reusing assets. Storage is not an issue, because individuals keep track of and distribute their own software assets. For such an ad hoc, individually driven approach, it is difficult to measure any potential benefits to the company.
- **facilitated reuse**. With this approach, developers are not required to practice reuse, but they are **encouraged** to do so.
- The organization makes available some tools and techniques that enable the development and sharing of reusable assets.
- Little is done to track the quality and use of reusable assets.
- **Managed Reuse** is a more structured, and more expensive, mode of managing software reuse, the development, sharing, and adoption of reusable assets is mandated.
- **Designed Reuse**: The organization establishes processes and policies for ensuring that reuse is practiced and that the results are measured.

Approaches to Reuse

Approach	Reuse Level	Cost	Policies & Procedures
Ad hoc	None to low	Low	None
Facilitated	Low	Low	Developers are encouraged to reuse but are not required to do so.
Managed	Moderate	Moderate	Development, sharing, and adoption of reusable assets are mandated; organizational policies are established for documentation, packaging, and certification.
Designed	High	High	Reuse is mandated; policies are put in place so that reuse effectiveness can be measured; code must be designed for reuse during initial development, regardless of the application it is originally designed for; there may be a corporate office for reuse.

Source: Based on Griss, 2003.